

Midpoint Transmutation, Dyadic Cardinal Reproduction, and Holonomic Obstructions

New deductions around the Fabius function, its inverse, Rvachev's up-function, Thue–Morse cancellation, Lambert–W asymptotics, and q-scale algebra

A corpus-relative frontier report for Vladimir Reshetnikov's ProveIt repository

Repository snapshot audited: 27 August 2026

Abstract

A recursive audit of the fifty-seven $\text{\LaTeX}$ sources under `Analysis/FabiusFunction/docs` reveals a mature body of work on exact dyadic arithmetic, Thue–Morse prefix sums, q-binomial coefficient extraction, infinite sinc and cosine products, sharp Fourier and endpoint asymptotics, lower-branch Lambert–W coordinates, Bell-polynomial saddle algebra, Lagrange interpolation, and Richardson acceleration. This report deliberately avoids reproducing those developments. Instead it develops three complementary branches that were not found in the audited corpus.

First, an exact identity transports the left-endpoint germ of the Fabius function to its midpoint:

$$F\left(\frac{1}{2} \pm h\right) = \frac{1}{2} \pm 2h \mp F(h), \quad 0 \leq h \leq \frac{1}{2}.$$

It follows that the Taylor jet of F at $1/2$ is exactly affine although F is not analytic there. For the inverse $I = F^{-1}$, writing $z = \delta/2$ and $h = I(1/2 + \delta) - 1/2$ gives the exact fixed-point equation $h = z + F(h)/2$. The inverse defect $E(\delta) = h - z$ is flat to every order and satisfies $E(\delta) \sim F(\delta/2)/2$. Therefore the repository's complete lower-Lambert endpoint expansion transfers, coefficient for coefficient, to the first nontrivial correction of the inverse at its midpoint. A finite, rigorous Lagrange–Bell jet with an integral remainder is also obtained.

Second, dyadic dilations $U_r(x) = 2^{-r} \text{up}(2^{-r}x)$ form a cardinal reproduction ladder. Their Fourier transforms have zeros of order at least $r+1$ at every nonzero integer, a property whose finite block is exactly the Prouhet cancellation of a length- 2^{r+1} signed Thue–Morse word. Consequently

$$\sum_{k \in \mathbb{Z}} (x - k)^j U_r(x - k) = \int_{\mathbb{R}} y^j U_r(y) dy, \quad 0 \leq j \leq r.$$

Reciprocal-moment Appell polynomials $\mathcal{A}_n^{[r]}$ then give finite exact formulas $\sum_k \mathcal{A}_n^{[r]}(k) U_r(x - k) = x^n$ for $n \leq r$. Their coefficients are explicit complete Bell polynomials in Bernoulli cumulants; their dyadic scale transport is governed by the Bernoulli-polynomial values $B_n(1/2)$; and, because $\mathcal{A}_n(x; Q)$ is polynomial in $Q = 4^r$, the scale sequence obeys exact q-binomial finite-difference closures at $q = 1/4$.

Third, two independent obstructions exclude holonomicity. The midpoint identity propagates nonanalyticity backward through the dyadic tent map, proving that F is nonanalytic at every dyadic rational; the same mechanism gives dense nonanalytic sets for F^{-1} and up . Hence none is locally D-finite on any open interval. Independently, the unbounded multiplicities $\text{ord}_m \widehat{\text{up}} = 1 + \nu_2(|m|)$ rule out a polynomial-coefficient differential equation for the entire Fourier transform. This implies non-D-finiteness of the moment generating functions, non-P-recursiveness of the moment sequences, and non-P-recursiveness of every fixed- x Appell coefficient sequence. Symbolic and high-precision numerical experiments accompany the proofs, and the fully commented Python program is included in the archive.

Priority and novelty disclaimer

The word *new* in this report means *not found in the audited repository snapshot*. It is not a claim of global historical priority. In particular, general Strang–Fix theory and nonstationary multiresolution analysis based on the up-function already exist in the approximation-theory literature [8, 7, 9]. The corpus-relative contribution here is the explicit dilation-indexed cardinal ladder, its Prouhet factorization, the Bernoulli–Bell coefficient formulas, and their exact q -scale closure, together with the midpoint/inverse and holonomic deductions.

Contents

1	Corpus audit and non-duplication protocol	4
1.1	What was audited	4
1.2	The overlap map	4
1.3	Status labels used below	5
2	Established framework and notation	5
2.1	The three functions	5
2.2	Probability, Fourier, and moments	5
2.3	The imported endpoint coordinate	6
3	An exact midpoint–endpoint transmutation	6
3.1	The complete midpoint jet	7
3.2	Central integral identities	7
4	The inverse midpoint defect and its Lambert–W transfer	8
4.1	An exact fixed-point equation	8
4.2	Certified monotone inversion	8
4.3	Sharp leading comparison	9
4.4	A rigorous Lagrange–Bell jet	9
5	Thue–Morse blocks and the dyadic zero architecture	10
5.1	Box convolutions	10
5.2	Exact integer multiplicities	11
6	A dyadic cardinal reproduction ladder	11
6.1	Strang–Fix vanishing at every scale	11
6.2	The sampling operator	12
6.3	Reciprocal-moment Appell correctors	12
7	Bernoulli cumulants and Bell-polynomial correctors	13
7.1	Closed cumulants	13
7.2	Complete Bell form	14
7.3	Bernoulli-polynomial dyadic transport	14
8	q-Lagrange interpolation and exact Richardson closure	15
8.1	Interpolation in the scale variable	15
8.2	Exact q -binomial annihilation	15

9	Dense nonanalyticity and holonomic obstructions	16
9.1	D-finite and P-recursive objects	16
9.2	Backward propagation through the tent map	16
9.3	The entire Fourier transform	17
9.4	Moments and Appell coefficients	17
10	Numerical and symbolic verification	18
10.1	Reproducible implementation	18
10.2	Midpoint identity and inverse defect	18
10.3	Exact dyadic arithmetic	19
10.4	Cardinal and q-scale checks	19
11	A formalization roadmap	20
11.1	Low-cost Lean targets	20
11.2	Finite combinatorial and algebraic targets	20
11.3	Fourier and periodization targets	20
11.4	Holonomic infrastructure	21
12	Further frontier problems	21
13	Conclusion	22
A	Additional proof details	22
A.1	Barycentric denominator calculation	22
A.2	Why the inverse correction is beyond every Lambert order	23
A.3	Ordinary-point analyticity	23
B	Explicit Appell polynomials through degree eight	23
C	Reproducibility files	23
D	Complete recursive TeX manifest	24

1 Corpus audit and non-duplication protocol

1.1 What was audited

The recursive tree under `Analysis/FabiusFunction/docs` contains twenty-five living or current-tree TeX sources and thirty-two frozen archive sources [1, 2, 3, 4]. The living group includes the primary exposition, the non-elementarity article, the Lean walkthrough, the mathematical glossary, the consolidated non-formalized frontier, the living Thue–Morse Formula Atlas, twelve Fourier-decay drafts or audits, and seven transcribed research papers. The archive preserves early notes, parallel primary expositions, supplements, q-calculus articles, walkthroughs, glossaries, per-topic frontier dossiers, and three predecessor Atlas consolidations.

The archive ledger explains that current-head byte-identical copies were pruned and that the remaining documents are frozen historical versions. Accordingly, this audit used three levels of comparison:

1. direct reading of the living canonical exposition, consolidated frontier, Atlas, distinct Fourier-decay documents, and paper transcriptions;
2. topic-by-topic comparison with every archived entry and its stated scope;
3. targeted full-text searches for the candidate branches developed below, including close variants of “Strang–Fix”, “polynomial reproduction”, “inverse-moment Appell”, “D-finite”, “holonomic”, “P-recursive”, and the continuous midpoint identity of [theorem 3.1](#).

The complete path inventory is reproduced in [section D](#) and included as a separate machine-readable file, `corpus_manifest.txt`.

1.2 The overlap map

The repository already treats the following subjects in substantial depth:

- exact rational values $F(a/2^n)$, triangular recurrences, composition sums, q-Pochhammer symbols, Gaussian binomials, and coefficient extraction;
- Thue–Morse product identities, Prouhet cancellation, iterated prefix sums, finite convolution laws, and sharp convergence rates;
- the sinc and cosine products for $\widehat{\text{up}}$, Fourier decay, zero sets, and refined log-periodic asymptotics;
- lower-branch Lambert–W coordinates for $F(x)$ as $x \downarrow 0$, periodic saddle jets, and all-orders Bell-polynomial contraction formulas;
- exact and asymptotic Lagrange interpolation, q-interpolation, and Richardson-type acceleration at dyadic scales;
- repeated-integration and finite-box-spline approximants to up , including exact tail replacement and quantitative error decompositions;
- inverse-function basics, including $I(1/2) = 1/2$, $I'(1/2) = 1/2$, and $I''(1/2) = 0$.

These are inputs, not claimed results of this report. In particular, the q-binomial weights in [section 8](#) are familiar from the repository’s q-Richardson work; what is new here is that they annihilate a newly constructed, exact polynomial-in-scale Appell family rather than an asymptotic error model.

Three principal corpus gaps

The audit did not reveal a treatment of:

1. the exact continuous midpoint–endpoint identity, its all-orders inverse consequence, or the transfer of endpoint Lambert–W asymptotics to the inverse midpoint defect;

2. a Strang–Fix/cardinal polynomial reproduction ladder for the dyadically dilated up-function, with reciprocal-moment Appell correctors and Bernoulli–Bell scale algebra;
3. D-finite and P-recursive obstructions derived simultaneously from dense dyadic nonanalyticity and unbounded Fourier-zero multiplicity.

1.3 Status labels used below

Gray boxes mark established inputs from the repository or standard literature. Green boxes mark deductions that appear new relative to the audited corpus. Orange boxes contain conjectures, extensions, or research programs whose proofs are not supplied. Numerical evidence is never used in place of proof.

2 Established framework and notation

2.1 The three functions

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius distribution function, normalized by

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1),$$

and, on $[0, 1]$, by

$$F'(x) = 2F(2x) \quad \left(0 \leq x \leq \frac{1}{2}\right), \quad F(1-x) = 1 - F(x). \quad (1)$$

It is C^∞ , strictly increasing on $(0, 1)$, and flat at both endpoints. Let $I = F^{-1} : (0, 1) \rightarrow (0, 1)$ be its ordinary inverse.

Rvachev’s even up-function $U = \text{up}$ is supported on $[-1, 1]$, integrates to one, and satisfies [5, 6]

$$U(x) = F(1 - |x|), \quad |x| \leq 1. \quad (2)$$

Equivalently, $F(x) = U(x - 1)$ on $[0, 1]$.

It is useful to write the symmetric tent map

$$\tau(x) = \begin{cases} 2x, & 0 \leq x \leq \frac{1}{2}, \\ 2 - 2x, & \frac{1}{2} \leq x \leq 1. \end{cases} \quad (3)$$

Differentiating the symmetry in (1) gives the global interior identity

$$F'(x) = 2F(\tau(x)), \quad 0 < x < 1. \quad (4)$$

2.2 Probability, Fourier, and moments

Use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx. \quad (5)$$

Let V_n be independent uniform variables on $[-2^{-n-1}, 2^{-n-1}]$. Then

$$X = \sum_{n=0}^{\infty} V_n \quad (6)$$

converges absolutely and has density U . Therefore

$$\widehat{U}(\xi) = \prod_{n=0}^{\infty} \frac{\sin(\pi\xi/2^n)}{\pi\xi/2^n} = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^n}\right), \quad (7)$$

$$= \prod_{m=1}^{\infty} \left(\cos \frac{\pi\xi}{2^m}\right)^m. \quad (8)$$

The moment generating function is the entire even function

$$\mathcal{M}_0(t) = \mathbb{E}e^{tX} = \prod_{n=0}^{\infty} \frac{\sinh(t/2^{n+1})}{t/2^{n+1}} = \widehat{U}\left(\frac{it}{2\pi}\right). \quad (9)$$

2.3 The imported endpoint coordinate

Put $L = \log 2$ and define the lower-Lambert coordinate [11, 3]

$$\lambda(x) = -\frac{1}{L}W_{-1}(-Lx), \quad \lambda(x)2^{-\lambda(x)} = x. \quad (10)$$

The repository develops a complete endpoint expansion whose first structural layer has the form

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(\lambda^{-N}). \quad (11)$$

Here $\lambda = \lambda(x)$; Ψ and the higher coefficient functions encode the dyadic phase and are periodic in the normalization of the frontier documents. The precise normalization of C_F , Ψ , and A_j is left to those sources. The point of [section 4](#) is that every term in (11) transfers unchanged to an inverse-function defect after replacing x by $\delta/2$ and subtracting $\log 2$.

Established input

Nothing in [section 2](#) is claimed as new. These identities fix conventions and isolate the exact inputs used later.

3 An exact midpoint–endpoint transmutation

Theorem 3.1 (Midpoint–endpoint transmutation). *For every $0 \leq h \leq 1/2$,*

$$F\left(\frac{1}{2} + h\right) = \frac{1}{2} + 2h - F(h), \quad (12)$$

$$F\left(\frac{1}{2} - h\right) = \frac{1}{2} - 2h + F(h). \quad (13)$$

Proof. Integrating the differential equation in (1) gives

$$F(h) = \int_0^h 2F(2t) dt = \int_0^{2h} F(u) du. \quad (14)$$

On the other hand,

$$\begin{aligned}
 F\left(\frac{1}{2}\right) - F\left(\frac{1}{2} - h\right) &= \int_{1/2-h}^{1/2} 2F(2t) \, dt \\
 &= \int_{1-2h}^1 F(u) \, du \\
 &= \int_0^{2h} F(1-v) \, dv \\
 &= 2h - \int_0^{2h} F(v) \, dv = 2h - F(h).
 \end{aligned}$$

Since $F(1/2) = 1/2$, this proves (13); symmetry gives (12). $\square$

Corpus-relative new deduction

The theorem says that the entire nontrivial midpoint germ is exactly the endpoint germ with a sign change after removal of the affine tangent. This is a functional identity, not the finite-approximant midpoint formula already present in the archived Thue–Morse convergence report.

3.1 The complete midpoint jet

Corollary 3.2 (Affine Taylor jet and nonanalyticity). *The derivatives of F at the midpoint are*

$$F\left(\frac{1}{2}\right) = \frac{1}{2}, \quad F'\left(\frac{1}{2}\right) = 2, \quad F^{(m)}\left(\frac{1}{2}\right) = 0 \quad (m \geq 2). \quad (15)$$

Nevertheless, F is not real analytic at $1/2$.

Proof. The endpoint flatness $F^{(m)}(0) = 0$ for every $m \geq 0$ and (12) give (15). If F were analytic at $1/2$, its Taylor series would force $F(1/2 + h) = 1/2 + 2h$ for all sufficiently small h , contradicting $F(h) > 0$ in (12). $\square$

Thus the flat endpoint term is invisible to every finite-order midpoint Taylor method. In numerical analysis language, it is a beyond-all-orders defect; in quasianalyticity language, the midpoint germ lies outside every class in which the Taylor map is injective.

3.2 Central integral identities

The transmutation has immediate exact integral consequences. For $0 \leq a \leq 1/2$,

$$\int_{1/2-a}^{1/2+a} F(x) \, dx = a, \quad (16)$$

$$\int_0^a \left[F\left(\frac{1}{2} + h\right) - \frac{1}{2} - 2h \right] \, dh = - \int_0^a F(h) \, dh. \quad (17)$$

The first is also a direct consequence of symmetry, while the second identifies the exact signed central curvature budget with the endpoint mass. Repeated integration of (12) similarly transports all endpoint primitives to the midpoint.

4 The inverse midpoint defect and its Lambert–W transfer

4.1 An exact fixed-point equation

Let $I = F^{-1}$. For $0 \leq \delta < 1/2$ define

$$h(\delta) = I\left(\frac{1}{2} + \delta\right) - \frac{1}{2}, \quad z = \frac{\delta}{2}, \quad E(\delta) = h(\delta) - z. \quad (18)$$

Since $F(1/2 + h) = 1/2 + \delta$, [theorem 3.1](#) gives the exact equation

$$\delta = 2h - F(h), \quad h = z + \frac{1}{2}F(h), \quad E(\delta) = \frac{1}{2}F(z + E(\delta)). \quad (19)$$

For negative δ , inverse symmetry gives

$$I\left(\frac{1}{2} + \delta\right) = \frac{1}{2} + \frac{\delta}{2} + \operatorname{sgn}(\delta)E(|\delta|). \quad (20)$$

Theorem 4.1 (All-orders inverse midpoint flatness). *The inverse I is C^∞ on $(0, 1)$ and*

$$I'\left(\frac{1}{2}\right) = \frac{1}{2}, \quad I^{(m)}\left(\frac{1}{2}\right) = 0 \quad (m \geq 2). \quad (21)$$

More strongly, for every $N \geq 1$,

$$E(\delta) = \mathcal{O}(\delta^N) \quad (\delta \downarrow 0). \quad (22)$$

The inverse is not analytic at $1/2$.

Proof. The inverse is smooth because $F'(x) > 0$ for $0 < x < 1$. From (19), $h \rightarrow 0$ with δ and

$$\frac{\delta}{h} = 2 - \frac{F(h)}{h} \rightarrow 2$$

because F is flat at zero. Hence $h \sim \delta/2$. The same equation yields $E = F(h)/2 = \mathcal{O}(h^N) = \mathcal{O}(\delta^N)$ for every N . Equation (20) then gives the complete jet (21). Analyticity would force $I(1/2 + \delta) = 1/2 + \delta/2$ locally, contradicting $E(\delta) > 0$. $\square$

4.2 Certified monotone inversion

The fixed-point equation is not merely asymptotic; it gives a certified algorithm.

Proposition 4.2 (Two-sided fixed-point iteration). *Fix $0 < \delta < 1/2$, put $z = \delta/2$, and define*

$$T_z(t) = z + \frac{1}{2}F(t), \quad b = z + \frac{1}{4}.$$

Then T_z maps $[z, b]$ into itself and is a contraction there with constant

$$q_\delta = F\left(\frac{1}{2} + \delta\right) < 1. \quad (23)$$

The lower iteration $\ell_0 = z, \ell_{n+1} = T_z(\ell_n)$ increases to h , while the upper iteration $u_0 = b, u_{n+1} = T_z(u_n)$ decreases to h . Moreover,

$$0 \leq h - \ell_n \leq \frac{q_\delta^n}{1 - q_\delta}(\ell_1 - \ell_0), \quad 0 \leq u_n - h \leq \frac{q_\delta^n}{1 - q_\delta}(u_0 - u_1). \quad (24)$$

Proof. For $t \in [z, b] \subset [0, 1/2)$, monotonicity gives $0 \leq F(t) \leq 1/2$, hence $z \leq T_z(t) \leq z + 1/4 = b$. Also

$$T'_z(t) = \frac{1}{2}F'(t) = F(2t) \leq F(2b) = F\left(\frac{1}{2} + \delta\right) = q_\delta < 1.$$

The contraction theorem and monotonicity prove the assertions; the displayed estimates are the standard geometric a posteriori bounds. $\square$

This gives a simple inverse evaluator requiring only forward evaluations of F , with no numerical differentiation and a rigorous enclosure at every step.

4.3 Sharp leading comparison

Proposition 4.3 (Endpoint proxy for the inverse defect). *For each $0 < \delta < 1/2$ there exists $\xi_\delta \in (z, h)$ such that*

$$E(\delta) = \frac{F(z)}{2 - F'(\xi_\delta)} = \frac{F(z)}{2} \frac{1}{1 - F'(\xi_\delta)/2}. \quad (25)$$

Consequently

$$E(\delta) \sim \frac{1}{2} F\left(\frac{\delta}{2}\right), \quad \delta \downarrow 0, \quad (26)$$

and

$$\log E(\delta) = \log F\left(\frac{\delta}{2}\right) - \log 2 + \mathcal{O}(F'(h(\delta))). \quad (27)$$

Proof. Apply the mean-value theorem to $F(z + E) - F(z)$ in (19):

$$2E = F(z) + F'(\xi_\delta)E.$$

This is (25). Since $\xi_\delta \rightarrow 0$ and $F'(0) = 0$, the ratio in (26) tends to one. Taking logarithms and using $-\log(1 - u) = \mathcal{O}(u)$ proves (27). $\square$

Lambert–W transfer theorem

Let $z = \delta/2$ and $\lambda = \lambda(z)$ as in (10). The correction term in (27) is smaller than every algebraic power of $1/\lambda$. Therefore any complete endpoint expansion of the form (11) transfers term by term:

$$\begin{aligned} \log E(\delta) = & -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) - \log 2 \\ & + \sum_{j=1}^{N-1} \frac{A_j(\lambda)}{\lambda^j} + \mathcal{O}(\lambda^{-N}). \end{aligned} \quad (28)$$

Thus the inverse midpoint defect has the same dyadic phase, the same Bell-polynomial saddle coefficients, and the same lower-Lambert coordinate as the endpoint function, evaluated at half the ordinate displacement.

4.4 A rigorous Lagrange–Bell jet

Set $\phi = F/2$ and introduce a coupling parameter ε :

$$h_\varepsilon = z + \varepsilon\phi(h_\varepsilon), \quad 0 \leq \varepsilon \leq 1. \quad (29)$$

For fixed $z \in (0, 1/4)$, the implicit-function denominator $1 - \varepsilon\phi'(h_\varepsilon)$ stays positive, so h_ε is C^∞ in ε .

Theorem 4.4 (Finite Lagrange–Bell expansion with exact remainder). *For every $N \geq 1$,*

$$h_1 = z + \sum_{n=1}^N \frac{1}{n!} D_z^{n-1}[\phi(z)^n] + R_N(z), \quad (30)$$

where

$$R_N(z) = \frac{1}{N!} \int_0^1 (1 - \varepsilon)^N \frac{\partial^{N+1} h_\varepsilon}{\partial \varepsilon^{N+1}} d\varepsilon. \quad (31)$$

Moreover,

$$D^{n-1}(\phi^n) = \sum_{k=0}^{n-1} (n)_k \phi^{n-k} B_{n-1,k}(\phi', \phi'', \dots), \quad (32)$$

where $B_{m,k}$ is the partial exponential Bell polynomial and $(n)_k$ is a falling factorial.

Proof. The Lagrange–Buerhmann coefficient identity

$$\left. \frac{\partial^n h_\varepsilon}{\partial \varepsilon^n} \right|_{\varepsilon=0} = D_z^{n-1} [\phi(z)^n]$$

follows by formal differentiation of (29); because only a finite jet is asserted, C^∞ regularity suffices and no analyticity of F is needed. Taylor’s theorem in ε with integral remainder gives (30)–(31). Formula (32) is Faà di Bruno’s formula applied to the power map $u \mapsto u^n$; see also the standard Bell-polynomial and umbral treatments [13, 12]. $\square$

The first four correction layers, evaluated formally at $\varepsilon = 1$, are

$$\begin{aligned} E = & \frac{F}{2} + \frac{FF'}{4} + \frac{F(F')^2}{8} + \frac{F^2F''}{16} \\ & + \frac{F(F')^3}{16} + \frac{3F^2F'F''}{32} + \frac{F^3F'''}{96} + \cdots, \end{aligned} \quad (33)$$

where every F and derivative is evaluated at $z = \delta/2$. Equation (30), rather than the ellipsis, is the rigorous statement. Whether the full coupling series converges at $\varepsilon = 1$ uniformly as $z \downarrow 0$ is left open.

5 Thue–Morse blocks and the dyadic zero architecture

5.1 Box convolutions

For $a > 0$ let

$$b_a(x) = \frac{1}{2a} \mathbf{1}_{[-a,a]}(x), \quad \widehat{b}_a(\xi) = \frac{\sin(2\pi a\xi)}{2\pi a\xi}. \quad (34)$$

Then

$$U = *_{n=0}^\infty b_{2^{-n-1}}. \quad (35)$$

For $r \geq 0$ define the normalized dilation

$$U_r(x) = 2^{-r} U(2^{-r}x), \quad \text{supp } U_r = [-2^r, 2^r], \quad \widehat{U}_r(\xi) = \widehat{U}(2^r\xi). \quad (36)$$

Its first $r + 1$ box factors form

$$B_r = *_{j=0}^r b_{2^{r-j-1}}. \quad (37)$$

Proposition 5.1 (Distributional Thue–Morse derivative). *In the sense of compactly supported distributions,*

$$D^{r+1} B_r = 2^{-r(r+1)/2} \sum_{n=0}^{2^{r+1}-1} (-1)^{\text{wt}_2(n)} \delta_{n-(2^{r+1}-1)/2}. \quad (38)$$

Proof. Since $Db_a = (\delta_{-a} - \delta_a)/(2a)$, convolving the $r + 1$ derivatives produces one atom for every binary choice of endpoints. If the chosen positive endpoints have binary indicator n , their signs multiply to $(-1)^{\text{wt}_2(n)}$, their location is

$$\sum_{j=0}^r (2\varepsilon_j - 1) 2^{r-j-1} = n - \frac{2^{r+1} - 1}{2},$$

and the product of normalizing factors is $\prod_{j=0}^r 2^{-(r-j)} = 2^{-r(r+1)/2}$. $\square$

The finite exponential sum in (38) has a zero of order $r + 1$ at every integer frequency, the classical Prouhet cancellation encoded by the Thue–Morse word [15, 14]. Indeed, after translation by an integer frequency, its Taylor coefficients through order r are the Prouhet sums

$$\sum_{n=0}^{2^{r+1}-1} (-1)^{\text{wt}_2(n)} n^j = 0, \quad 0 \leq j \leq r. \quad (39)$$

Thus the universal part of the integer zero multiplicity is literally the moment cancellation of one finite Thue–Morse block.

5.2 Exact integer multiplicities

Proposition 5.2 (Dyadic zero multiplicity). *For every nonzero integer m ,*

$$\text{ord}_m \widehat{U} = 1 + \nu_2(|m|). \quad (40)$$

Consequently, for every nonzero $\ell \in \mathbb{Z}$,

$$\text{ord}_\ell \widehat{U}_r = r + 1 + \nu_2(|\ell|). \quad (41)$$

Proof. In (7), the factor indexed by n vanishes at an integer m exactly when 2^n divides m . These zeros are simple, and all remaining factors are nonzero there, proving (40). Apply it to $2^r \ell$ to obtain (41). $\square$

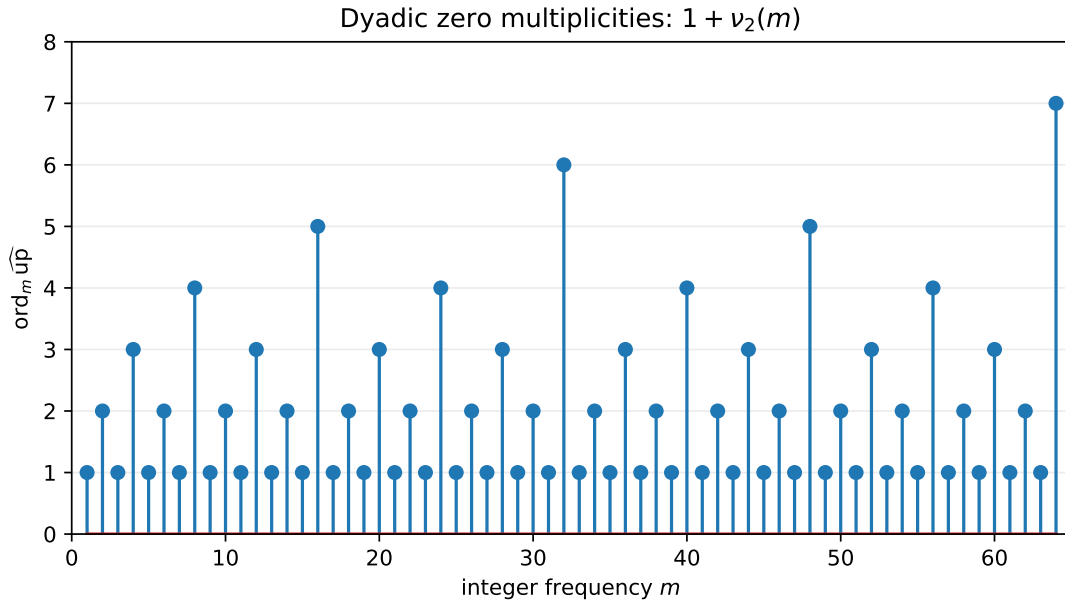


Figure 1: The exact multiplicities $1 + \nu_2(m)$ for $1 \leq m \leq 64$. The unbounded spikes at powers of two later obstruct D-finiteness of the entire Fourier transform.

Remark 5.3. The multiplicity formula is not used as a novelty claim by itself. Its role here is structural: the first $r + 1$ units of multiplicity are explained by the Prouhet block (38), while the remaining $\nu_2(\ell)$ units come from the infinite tail. This split is precisely what drives the cardinal reproduction order in the next section.

6 A dyadic cardinal reproduction ladder

6.1 Strang–Fix vanishing at every scale

Equation (41) implies

$$\widehat{U}_r^{(j)}(\ell) = 0, \quad \ell \in \mathbb{Z} \setminus \{0\}, \quad 0 \leq j \leq r. \quad (42)$$

This is the one-dimensional Strang–Fix condition of order $r + 1$ on the unit lattice. General approximation theory already connects such vanishing to polynomial reproduction [8, 7, 9]. The following exact formulas specialize that principle to the up-function and keep track of all moment corrections.

Theorem 6.1 (Moment periodization). *Let*

$$\mu_j^{[r]} = \int_{\mathbb{R}} y^j U_r(y) dy. \quad (43)$$

For every $r \geq 0$, every $0 \leq j \leq r$, and every $x \in \mathbb{R}$,

$$\sum_{k \in \mathbb{Z}} (x - k)^j U_r(x - k) = \mu_j^{[r]}. \quad (44)$$

The sum is finite, containing at most $2^{r+1} + 2$ nonzero terms.

Proof. The left side is a smooth 1-periodic function. Its ℓ th Fourier coefficient is

$$\begin{aligned} \int_0^1 \sum_k (x - k)^j U_r(x - k) e^{-2\pi i \ell x} dx &= \int_{\mathbb{R}} y^j U_r(y) e^{-2\pi i \ell y} dy \\ &= \frac{\widehat{U}_r^{(j)}(\ell)}{(-2\pi i)^j}. \end{aligned}$$

For $\ell \neq 0$ this vanishes by (42); at $\ell = 0$ it is $\mu_j^{[r]}$. Hence the periodic function is constant. $\square$

The first cases are particularly transparent:

$$\sum_k U(x - k) = 1, \quad (45)$$

$$\sum_k (x - k) U_1(x - k) = 0, \quad (46)$$

$$\sum_k (x - k)^2 U_2(x - k) = \frac{16}{9}. \quad (47)$$

6.2 The sampling operator

For a polynomial p , define

$$(Q_r p)(x) = \sum_{k \in \mathbb{Z}} p(k) U_r(x - k). \quad (48)$$

Let

$$\mathcal{M}_r(t) = \int_{\mathbb{R}} e^{ty} U_r(y) dy = \sum_{j=0}^{\infty} \mu_j^{[r]} \frac{t^j}{j!} = \mathcal{M}_0(2^r t). \quad (49)$$

Proposition 6.2 (Exact differential action on low-degree polynomials). *If $\deg p \leq r$, then*

$$Q_r p = \mathcal{M}_r(-D)p = \mathcal{M}_r(D)p. \quad (50)$$

Proof. Taylor-expand $p(k) = p(x - (x - k))$ and use [theorem 6.1](#):

$$Q_r p(x) = \sum_{j=0}^{\deg p} \frac{(-1)^j}{j!} p^{(j)}(x) \sum_k (x - k)^j U_r(x - k) = \sum_j \mu_j^{[r]} \frac{(-D)^j}{j!} p(x).$$

The distribution U_r is even, so all odd moments vanish and $\mathcal{M}_r(-D) = \mathcal{M}_r(D)$. $\square$

6.3 Reciprocal-moment Appell correctors

Define monic polynomials $\mathcal{A}_n^{[r]}(x)$ by

$$\sum_{n=0}^{\infty} \mathcal{A}_n^{[r]}(x) \frac{t^n}{n!} = \frac{e^{xt}}{\mathcal{M}_r(t)}. \quad (51)$$

They form an Appell sequence:

$$\frac{d}{dx} \mathcal{A}_n^{[r]}(x) = n \mathcal{A}_{n-1}^{[r]}(x). \quad (52)$$

They are distinct from the q -Appell blocks occurring in the repository's exact dyadic endpoint formulas; the present family is defined by reciprocal moments of a dilated density.

Theorem 6.3 (Finite exact cardinal reproduction). *For every $r \geq 0$ and every $0 \leq n \leq r$,*

$$\sum_{k \in \mathbb{Z}} \mathcal{A}_n^{[r]}(k) U_r(x - k) = x^n, \quad x \in \mathbb{R}. \quad (53)$$

Only finitely many terms are nonzero.

Proof. By [theorem 6.2](#), applying Q_r to the generating polynomial through degree r is the same as applying $\mathcal{M}_r(-D)$. Since $\mathcal{M}_r$ is even,

$$\mathcal{M}_r(-D) \frac{e^{xt}}{\mathcal{M}_r(t)} = \frac{\mathcal{M}_r(-t)}{\mathcal{M}_r(t)} e^{xt} = e^{xt}.$$

Coefficient comparison gives (53) for $n \leq r$. □

Example 6.4 (The first exact reconstruction identities). For all real x ,

$$\sum_k U(x - k) = 1, \quad (54)$$

$$\sum_k k U_1(x - k) = x, \quad (55)$$

$$\sum_k \left(k^2 - \frac{16}{9} \right) U_2(x - k) = x^2, \quad (56)$$

$$\sum_k \left(k^3 - \frac{64}{3} k \right) U_3(x - k) = x^3. \quad (57)$$

Each identity is finite at every x because $U_r(x - k) = 0$ when $|x - k| > 2^r$.

Corpus-relative new deduction

The dilation direction is the reverse of a standard shrinking-mesh approximation scheme: the support grows with r , while the fixed unit lattice sees progressively higher integer-frequency zero multiplicity. This “coarsening reproduction ladder” is therefore best viewed as an exact cardinal identity or moment-extraction device, not by itself as a stable numerical approximation method. Stability and conditioning are separate questions, discussed in [section 12](#).

7 Bernoulli cumulants and Bell-polynomial correctors

7.1 Closed cumulants

The classical expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m} \quad (58)$$

and (9) give

$$\log \mathcal{M}_0(t) = \sum_{m=1}^{\infty} \kappa_{2m} \frac{t^{2m}}{(2m)!}, \quad \kappa_{2m} = \frac{2^{2m-1} B_{2m}}{m(2^{2m} - 1)}, \quad \kappa_{2m+1} = 0. \quad (59)$$

At scale r ,

$$\kappa_{2m}^{[r]} = 2^{2mr} \kappa_{2m} = Q^m \kappa_{2m}, \quad Q = 4^r. \quad (60)$$

The first values are

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}, \quad \kappa_{10} = \frac{256}{33759}. \quad (61)$$

7.2 Complete Bell form

Let Y_n denote the complete exponential Bell polynomial, characterized by

$$\exp\left(\sum_{j \geq 1} a_j \frac{t^j}{j!}\right) = \sum_{n \geq 0} Y_n(a_1, \dots, a_n) \frac{t^n}{n!}.$$

From (51) and (60),

$$\boxed{\mathcal{A}_n(x; Q) = Y_n(x, -\kappa_2 Q, 0, -\kappa_4 Q^2, 0, -\kappa_6 Q^3, \dots), \quad \mathcal{A}_n^{[r]}(x) = \mathcal{A}_n(x; 4^r).} \quad (62)$$

Thus, for fixed n , $\mathcal{A}_n(x; Q)$ is a polynomial in Q of degree $\lfloor n/2 \rfloor$. This elementary observation is the source of the exact q -scale algebra in [section 8](#).

The first polynomials are

$$\mathcal{A}_0(x; Q) = 1, \quad (63)$$

$$\mathcal{A}_1(x; Q) = x, \quad (64)$$

$$\mathcal{A}_2(x; Q) = x^2 - \frac{Q}{9}, \quad (65)$$

$$\mathcal{A}_3(x; Q) = x^3 - \frac{Q}{3}x, \quad (66)$$

$$\mathcal{A}_4(x; Q) = x^4 - \frac{2}{3}Qx^2 + \frac{31}{675}Q^2, \quad (67)$$

$$\mathcal{A}_5(x; Q) = x^5 - \frac{10}{9}Qx^3 + \frac{31}{135}Q^2x, \quad (68)$$

$$\mathcal{A}_6(x; Q) = x^6 - \frac{5}{3}Qx^4 + \frac{217}{315}Q^2x^2 - \frac{2347}{59535}Q^3. \quad (69)$$

Higher formulas through degree eight are listed in [section B](#) and generated exactly by the accompanying Python program.

7.3 Bernoulli-polynomial dyadic transport

The functional equation for the moment generating functions is

$$\mathcal{M}_{r+1}(t) = \frac{\sinh(2^r t)}{2^r t} \mathcal{M}_r(t). \quad (70)$$

The Bernoulli-polynomial generating function gives

$$\frac{z}{\sinh z} = \frac{2ze^z}{e^{2z} - 1} = \sum_{k=0}^{\infty} 2^k B_k\left(\frac{1}{2}\right) \frac{z^k}{k!}. \quad (71)$$

All odd terms vanish. For $k = 2j$,

$$c_{2j} := 2^{2j} B_{2j}\left(\frac{1}{2}\right) = 2(1 - 2^{2j-1}) B_{2j}. \quad (72)$$

Multiplying generating functions yields an exact scale recurrence.

Theorem 7.1 (Bernoulli transport across one dyadic dilation). *For every $n, r \geq 0$,*

$$\mathcal{A}_n^{[r+1]}(x) = \sum_{k=0}^n \binom{n}{k} 2^{(r+1)k} B_k\left(\frac{1}{2}\right) \mathcal{A}_{n-k}^{[r]}(x) \quad (73)$$

$$= \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} c_{2j} 2^{2jr} \mathcal{A}_{n-2j}^{[r]}(x). \quad (74)$$

Proof. Divide (51) at scale $r + 1$ by its counterpart at scale r and use (70):

$$\frac{e^{xt}}{\mathcal{M}_{r+1}(t)} = \frac{2^r t}{\sinh(2^r t)} \frac{e^{xt}}{\mathcal{M}_r(t)}.$$

Insert (71) with $z = 2^r t$ and compare coefficients. $\square$

This identity gives a direct, exact connection among the up-function, Bernoulli polynomials, Bell polynomials, and dyadic scaling. It also supplies an efficient symbolic construction: one can build the entire scale tower without re-expanding the infinite product.

8 q -Lagrange interpolation and exact Richardson closure

8.1 Interpolation in the scale variable

Fix n and put $d = \lfloor n/2 \rfloor$. Since $\mathcal{A}_n(x; Q)$ has degree at most d in Q , any $d + 1$ distinct scales determine it exactly. Let

$$q = \frac{1}{4}, \quad Q_r = 4^r = q^{-r}. \quad (75)$$

Then ordinary Lagrange interpolation gives

$$\mathcal{A}_n(x; Q) = \sum_{j=0}^d \mathcal{A}_n^{[j]}(x) L_j(Q), \quad L_j(Q) = \prod_{\substack{0 \leq \ell \leq d \\ \ell \neq j}} \frac{Q - q^{-\ell}}{q^{-j} - q^{-\ell}}. \quad (76)$$

The barycentric denominators have the closed q -binomial form

$$\begin{aligned} \frac{1}{\prod_{\ell \neq j} (q^{-j} - q^{-\ell})} &= \frac{(-1)^{d-j} q^{\binom{d+1}{2} + \binom{j}{2}}}{(q; q)_j (q; q)_{d-j}} \\ &= \frac{(-1)^{d-j} q^{\binom{d+1}{2} + \binom{j}{2}}}{(q; q)_d} \begin{bmatrix} d \\ j \end{bmatrix}_q. \end{aligned} \quad (77)$$

Thus the scale reconstruction is naturally expressed in the same Gaussian-binomial language as the repository's exact dyadic arithmetic.

8.2 Exact q -binomial annihilation

Theorem 8.1 (Finite q -scale closure). *Let $N = d + 1 = \lfloor n/2 \rfloor + 1$. For every $r \geq 0$,*

$$\sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q \mathcal{A}_n^{[r+j]}(x) = 0, \quad q = \frac{1}{4}. \quad (78)$$

The identity is polynomial in x and exact over $\mathbb{Q}$.

Proof. It suffices to test a monomial Q^m with $0 \leq m < N$. At the shifted nodes $Q_{r+j} = q^{-(r+j)}$, the left side becomes

$$q^{-mr} \sum_{j=0}^N (-1)^j q^{\binom{j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_q (q^{-m})^j = q^{-mr} (q^{-m}; q)_N.$$

The displayed equality is the finite q -binomial theorem, and one factor in $(q^{-m}; q)_N$ is $1 - q^{-m} q^m = 0$. $\square$

Example 8.2. For $n = 4$ or $n = 5$, $d = 2$ and $N = 3$. Clearing denominators in (78) gives

$$64\mathcal{A}_n^{[r]} - 84\mathcal{A}_n^{[r+1]} + 21\mathcal{A}_n^{[r+2]} - \mathcal{A}_n^{[r+3]} = 0. \quad (79)$$

For $n = 6$ or 7 , the corresponding integer weights are 4096, -5440 , 1428, -85 , 1.

Exact closure, not asymptotic acceleration

Richardson extrapolation normally annihilates the first terms of an *asymptotic* scale expansion. Here the same q-binomial algebra annihilates the sequence exactly because the new Appell corrector is a finite polynomial in 4^r . This is a structural closure law and a consistency test for symbolic or numerical implementations. It should not be confused with the q-Richardson acceleration schemes already developed in the repository.

9 Dense nonanalyticity and holonomic obstructions

9.1 D-finite and P-recursive objects

Following the standard holonomic terminology [10], a function analytic near a point is D-finite if it satisfies a nontrivial linear differential equation

$$p_m(z)y^{(m)}(z) + \cdots + p_1(z)y'(z) + p_0(z)y(z) = 0, \quad p_j \in \mathbb{C}[z]. \quad (80)$$

At any ordinary point z_0 with $p_m(z_0) \neq 0$, every solution is analytic and is uniquely determined by its first m initial derivatives. A sequence is P-recursive if it satisfies a polynomial-coefficient recurrence; equivalently, its ordinary or exponential generating function is D-finite after the standard factorial normalization.

We use two independent obstructions:

1. a physical-space obstruction: dense sets of nonanalytic points;
2. a Fourier-space obstruction: zeros of unbounded multiplicity at ordinary points.

9.2 Backward propagation through the tent map

Proposition 9.1 (Dyadic nonanalyticity propagation). *The Fabius function is not analytic at any dyadic rational in $(0, 1)$.*

Proof. Nonanalyticity at $1/2$ is [theorem 3.2](#). Let $x_0 \neq 1/2$. If F were analytic near x_0 , then F' would be analytic there; by (4), $F \circ \tau = F'/2$ would be analytic there. Since τ is locally affine and invertible away from $1/2$, F would be analytic near $\tau(x_0)$. Contrapositively, nonanalyticity at $\tau(x_0)$ forces nonanalyticity at x_0 .

If $x_0 = k/2^m$ is in lowest dyadic terms, each application of τ lowers the denominator exponent by one and preserves an odd numerator. After finitely many steps it reaches $1/2$. Backward propagation completes the proof. $\square$

The dyadic rationals are dense. Therefore F has a dense set of nonanalytic points even though it is C^∞ everywhere.

Theorem 9.2 (Nowhere local holonomicity of F , I , and U). *No nontrivial polynomial-coefficient linear differential equation can hold for F on any nonempty open subinterval of $(0, 1)$. The same is true for the inverse $I = F^{-1}$ on every open subinterval of $(0, 1)$ and for $U = \text{up}$ on every open subinterval of $(-1, 1)$.*

Proof. Suppose (80) held for F on an open interval J . The leading polynomial has only finitely many zeros, so one can choose a dyadic $x_0 \in J$ at which $p_m(x_0) \neq 0$. Ordinary-point theory would make F analytic near x_0 , contradicting [theorem 9.1](#).

For the inverse, let $\mathbb{D}_2$ be the dyadics in $(0, 1)$. Since F is a homeomorphism, $F(\mathbb{D}_2)$ is dense. At $x_0 \in \mathbb{D}_2$, $F'(x_0) > 0$. If I were analytic near $y_0 = F(x_0)$, then $I'(y_0) = 1/F'(x_0) \neq 0$ and the analytic inverse theorem would make F analytic near x_0 , a contradiction. Thus I is nonanalytic on the dense set $F(\mathbb{D}_2)$, and the ordinary-point argument applies again.

Finally, $U(x) = F(x + 1)$ on $(-1, 0)$ and U is even. Translated and reflected dyadic sets are dense in $(-1, 1)$, giving the same conclusion. $\square$

Remark 9.3. The result is stronger than saying that the three functions are not elementary. D-finite functions include many non-elementary special functions, such as Bessel and Airy functions. The obstruction here excludes the entire holonomic class locally on every interval.

9.3 The entire Fourier transform

Theorem 9.4 (Unbounded zero multiplicity excludes D-finiteness). *The entire function $\widehat{U}$ is not D-finite over $\mathbb{C}(z)$.*

Proof. Assume that $\widehat{U}$ satisfies an equation of order m as in (80). Choose N so large that $N + 1 \geq m$ and 2^N is not a zero of the leading polynomial p_m . By [theorem 5.2](#), $\widehat{U}$ has a zero of multiplicity $N + 1$ at 2^N , so its first m initial derivatives vanish there. Uniqueness at the ordinary point 2^N would force $\widehat{U}$ to vanish identically, contrary to $\widehat{U}(0) = 1$. $\square$

The proof is robust: any nonzero entire function with zeros of unbounded multiplicity outside a finite exceptional set is non-D-finite.

9.4 Moments and Appell coefficients

Corollary 9.5 (Non-P-recursive moments). *For every $r \geq 0$, the moment generating function $\mathcal{M}_r$ is not D-finite. Hence the full moment sequence $\{\mu_n^{[r]}\}_{n \geq 0}$ is not P-recursive. The even subsequence $\{\mu_{2n}^{[r]}\}_{n \geq 0}$ is not P-recursive either.*

Proof. The linear substitution in (9) and dyadic scaling preserve D-finiteness, so D-finiteness of $\mathcal{M}_r$ would imply D-finiteness of $\widehat{U}$, contradicting [theorem 9.4](#). A P-recursive moment sequence would have a D-finite exponential generating function. For the even subsequence, suppose that $c_n = \mu_{2n}^{[r]}$ were P-recursive. Termwise multiplication by the hypergeometric sequence $1/(2n)!$ preserves P-recursiveness (equivalently, D-finite ordinary generating functions are closed under Hadamard products), so $H(z) = \sum_{n \geq 0} \mu_{2n}^{[r]} z^n / (2n)!$ would be D-finite. Since the odd moments vanish by symmetry, $\mathcal{M}_r(t) = H(t^2)$; closure under algebraic substitution would make $\mathcal{M}_r$ D-finite, again a contradiction. $\square$

Corollary 9.6 (Non-P-recursive reciprocal-moment Appell sequences). *Fix $r \geq 0$ and $x \in \mathbb{C}$. The sequence $\{\mathcal{A}_n^{[r]}(x)\}_{n \geq 0}$ is not P-recursive.*

Proof. Its exponential generating function is $G_{r,x}(t) = e^{xt} / \mathcal{M}_r(t)$. The zeros of $\widehat{U}_r$ at $m/2^r$, $m \in \mathbb{Z} \setminus \{0\}$, become zeros of $\mathcal{M}_r$ at

$$t = -\frac{2\pi i m}{2^r}. \quad (81)$$

Since the exponential numerator never vanishes, $G_{r,x}$ has infinitely many finite poles. A D-finite germ has finite singular set in the finite plane, contained in the zeros of the leading polynomial. Thus $G_{r,x}$ is not D-finite, and its coefficient sequence is not P-recursive. $\square$

Two-sided holonomic exclusion

The physical functions fail holonomicity because ordinary points would force analyticity where a dense dyadic set is nonanalytic. The Fourier and generating functions fail for the opposite reason: they are entire or meromorphic, but their zero/pole geometry is too rich for a finite-order polynomial differential equation. The two proofs are logically independent and illuminate different aspects of the same dyadic self-similarity.

10 Numerical and symbolic verification

10.1 Reproducible implementation

The archive contains `frontier_experiments.py`, a fully commented standalone program using `mpmath`, `SymPy`, `NumPy`, and `Matplotlib`. It performs the following tasks:

- evaluates $\widehat{U}$ from the sinc product and U from its period-two cosine series;
- checks the midpoint identity at high precision;
- solves the inverse fixed-point equation and compares $E(\delta)$ with $F(\delta/2)/2$;
- computes exact dyadic endpoint values from the triangular recurrence using rational arithmetic;
- constructs cumulants and Appell polynomials symbolically;
- verifies every q-closure through degree twelve exactly in $\mathbb{Q}[x, Q]$;
- tests cardinal reproduction at scales $0 \leq r \leq 5$;
- generates [figures 1](#) and [2](#) and CSV ledgers.

The run used 70 decimal digits, 420 half-integer Fourier modes, and 240 sinc factors per Fourier coefficient. Numerical residuals measure Fourier truncation and cancellation only; the identities themselves were proved above.

10.2 Midpoint identity and inverse defect

The largest absolute residual among $h = 0.05, 0.10, 0.20, 0.30, 0.45$ in (12) was 4.04×10^{-20} . Selected inverse-defect values are shown in [table 1](#).

Table 1: Inverse midpoint defect compared with its leading endpoint proxy.

δ	$E(\delta)$	$F(\delta/2)/2$	ratio
0.25	$1.86946480001 \times 10^{-3}$	$1.73611111111 \times 10^{-3}$	1.0768117248
0.20	$5.58337327435 \times 10^{-4}$	$5.41717809778 \times 10^{-4}$	1.0306792898
0.15	$1.06205195288 \times 10^{-4}$	$1.05314288462 \times 10^{-4}$	1.0084595057
0.10	$8.10084684889 \times 10^{-6}$	$8.09206623963 \times 10^{-6}$	1.0010850887
0.075	$1.09685028275 \times 10^{-6}$	$1.09661923462 \times 10^{-6}$	1.0002106913
0.05	$5.11439387869 \times 10^{-8}$	$5.11431110610 \times 10^{-8}$	1.0000161845

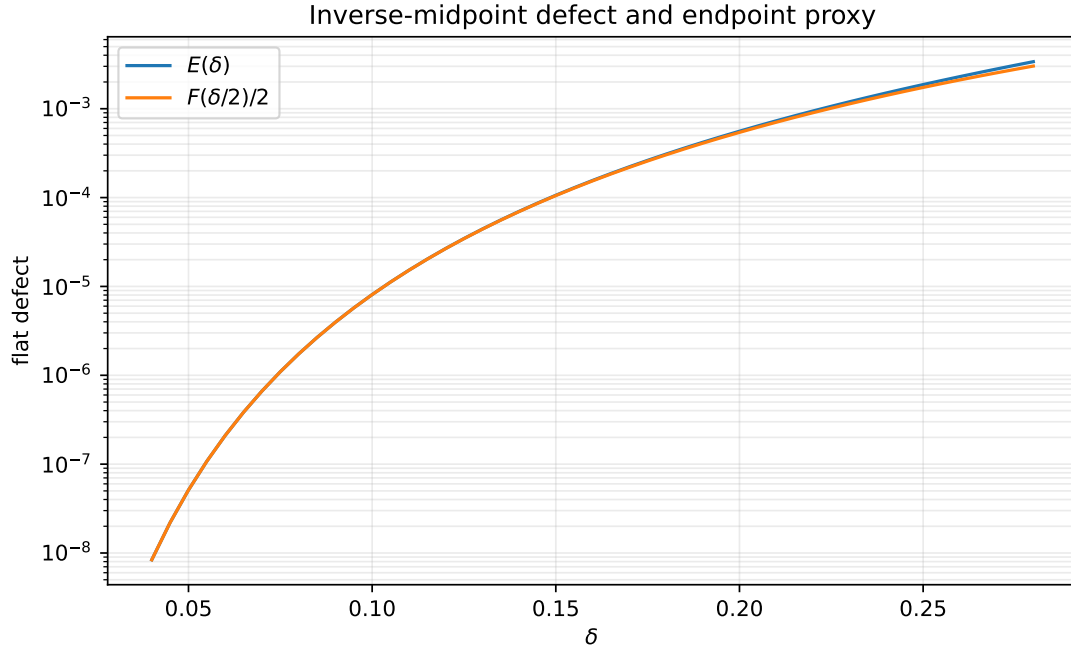


Figure 2: The exact inverse midpoint defect $E(\delta)$ and the leading endpoint proxy $F(\delta/2)/2$. Their relative separation rapidly becomes invisible on the logarithmic vertical scale, in agreement with [theorem 4.3](#).

10.3 Exact dyadic arithmetic

The code reproduces, using exact fractions,

$$\begin{aligned}
 F(1) &= 1, & F\left(\frac{1}{2}\right) &= \frac{1}{2}, & F\left(\frac{1}{4}\right) &= \frac{5}{72}, \\
 F\left(\frac{1}{8}\right) &= \frac{1}{288}, & F\left(\frac{1}{16}\right) &= \frac{143}{2073600}, & F\left(\frac{1}{32}\right) &= \frac{19}{33177600}, \\
 F\left(\frac{1}{64}\right) &= \frac{1153}{561842749440}.
 \end{aligned}$$

These values are not new; they serve as independent regression tests for the numerical evaluator and establish compatibility with the repository’s exact arithmetic.

10.4 Cardinal and q-scale checks

All q-binomial closures (78) through degree twelve simplified identically to zero. For the cardinal identities, the maximum residual over the five nonlattice points $-0.83, -0.41, 0.07, 0.37, 0.79$ is summarized in [table 2](#).

Table 2: Largest numerical residual in (53) at each tested scale. The final column reports the hardest case $n = r$.

r	degrees tested	maximum residual
0	0	7.25×10^{-71}
1	0, 1	4.52×10^{-20}
2	0, 1, 2	2.36×10^{-19}
3	0, 1, 2, 3	4.72×10^{-18}
4	0, 1, 2, 3, 4	5.67×10^{-17}
5	0, 1, 2, 3, 4, 5	5.23×10^{-14}

The growth in the last row is expected from cancellation between large polynomial samples and does not indicate failure of the exact identity. Increasing the Fourier truncation reduces the approximation error; interval arithmetic would instead provide a certified enclosure.

11 A formalization roadmap

The deductions split naturally into algebraic, analytic, and special-function layers.

11.1 Low-cost Lean targets

The following results require only the already formalized Fabius equations, elementary integration, and inverse monotonicity:

1. [theorem 3.1](#) and the all-orders derivative statement [theorem 3.2](#);
2. the inverse fixed-point equation (19);
3. flatness of E from endpoint flatness and $h \sim \delta/2$;
4. monotone lower and upper fixed-point iterations from [theorem 4.2](#).

A practical theorem statement should avoid asymptotic notation initially and prove, for every N , a neighborhood estimate $|E(\delta)| \leq C_N |\delta|^N$.

11.2 Finite combinatorial and algebraic targets

The distributional identity (38) can be formalized first as a finite signed-measure convolution identity. Its coefficient and location maps are binary sums, and the sign is exactly the existing signed Thue–Morse sequence. The q-closure (78) reduces to the finite q-binomial theorem and polynomial degree bounds. The Bernoulli transport recurrence is a formal power-series coefficient calculation and is likewise independent of analytic convergence.

11.3 Fourier and periodization targets

A formal proof of [theorem 6.1](#) needs:

- Fourier transforms of compactly supported smooth functions;
- differentiation under the integral;
- periodization and Fourier coefficients;
- the zero multiplicity of a finite or infinite sinc product.

A staged approach can first prove the identities for finite box convolutions, where all transforms are finite products, and then pass to U_r by the existing approximation limits.

11.4 Holonomic infrastructure

The D-finite results require a library definition of polynomial-coefficient linear differential equations and the ordinary-point analyticity/uniqueness theorem. Once that infrastructure exists, both obstructions are short:

- dense dyadic nonanalyticity contradicts ordinary-point analyticity;
- a zero of multiplicity at least the equation order at an ordinary point contradicts uniqueness.

The P-recursive corollaries then require the standard equivalence between recurrences and D-finite generating functions.

12 Further frontier problems

1. Differential transcendence

Non-D-finiteness excludes linear polynomial-coefficient differential equations but not nonlinear algebraic differential equations. Determine whether F , I , U , $\hat{U}$, or $\mathcal{M}_0$ is differentially algebraic over $\mathbb{C}(x)$. The infinite hierarchy of zero multiplicities and the dilation functional equations may provide a route to differential transcendence.

2. Convergence and resurgence of the inverse Lagrange–Bell coupling series

The finite jet (30) is rigorous, but convergence of the full series at $\varepsilon = 1$ is not asserted. Estimate the growth of $D^{n-1}[F(z)^n]$ uniformly as $z \downarrow 0$, identify a possible Gevrey order, and test Borel summability. The lower-Lambert coordinate should separate algebraic saddle corrections from the flat fixed-point feedback.

3. Exact two-adic denominator laws for the cardinal Appell family

The coefficients in (69) combine Bernoulli denominators with the factors $2^{2m} - 1$. Determine exact 2-adic and odd-prime valuations of the coefficients of $\mathcal{A}_n(x; Q)$, and compare them with the denominator phenomena in the repository’s exact dyadic Fabius values.

4. Stability of coarsening reproduction

The identities (53) are exact, but the sampling coefficients grow with r and n . Determine the optimal condition numbers, Lebesgue-type constants, and noise amplification of the finite cardinal sums. Seek dual kernels or mixed scales that preserve exactness while controlling cancellation.

5. Base- b Prouhet kernels

Replace the dyadic random series by uniform digits in base b and the Thue–Morse sign by a root-of-unity digit character. The finite block should produce a Prouhet cancellation of prescribed order, while the infinite tail produces a b -adic zero-multiplicity correction. Develop the analogue of the Appell ladder and identify its q -scale closure at $q = b^{-2}$.

6. q -holonomic versus P-recursive scale sequences

For fixed n , the scale sequence $r \mapsto \mathcal{A}_n^{[r]}(x)$ satisfies a finite q -binomial closure because it is a polynomial in 4^r . For fixed r , the degree sequence $n \mapsto \mathcal{A}_n^{[r]}(x)$ is not P-recursive. This sharp

asymmetry suggests studying whether it is q -holonomic in n , or whether the infinite pole lattice also rules out natural q -difference classes.

7. Phase-resolved inverse extrapolation

Equation (28) says that the inverse midpoint defect inherits the endpoint periodic phase exactly. Construct phase-locked Richardson schemes directly for $I(1/2 + \delta)$, and compare them with schemes that first approximate F and then invert. The fixed-point enclosure supplies rigorous error certificates.

8. Multivariate pseudo-box analogues

The approximation literature contains multivariate pseudo-box splines and nonstationary refinable systems. Determine which products possess a valuation-indexed zero multiplicity strong enough to create an increasing cardinal reproduction ladder, and whether multivariate Prouhet arrays replace the finite Thue–Morse block in (38).

13 Conclusion

The corpus already explains why q -binomial coefficients, infinite sinc products, Lambert–W functions, Bell polynomials, Lagrange interpolation, and Richardson extrapolation occur around the Fabius and Rvachev functions. The present report adds three cross-connections.

The midpoint is not merely symmetric: after subtracting its affine tangent, it is exactly the negative of the endpoint function. This identity upgrades the known first two inverse derivatives to an all-orders flatness theorem, furnishes a certified fixed-point inverse, and transports the complete endpoint Lambert–W expansion to the inverse midpoint defect.

The integer zeros of the Fourier transform are not merely spectral curiosities: under dyadic dilation, their multiplicities form a Strang–Fix ladder. A finite Prouhet–Thue–Morse block supplies the universal order, moment periodization supplies exact cardinal identities, reciprocal moments supply Appell correctors, Bernoulli cumulants and Bell polynomials supply their coefficients, and Gaussian binomials supply exact scale closures.

Finally, the same dyadic structure obstructs holonomicity in two directions. In physical space it produces dense nonanalyticity; in Fourier space it produces zeros of unbounded multiplicity. The consequences extend from F , F^{-1} , and up to their moment and Appell sequences. These deductions are compact enough to be plausible formalization targets, yet broad enough to open new research programs in differential algebra, exact sampling, q -holonomicity, and base- b generalization.

A Additional proof details

A.1 Barycentric denominator calculation

For completeness, let $0 \leq j \leq d$. Split the denominator into indices below and above j :

$$\begin{aligned} \prod_{\ell \neq j} (q^{-j} - q^{-\ell}) &= \left[\prod_{\ell=0}^{j-1} q^{-j} (1 - q^{j-\ell}) \right] \left[\prod_{\ell=j+1}^d -q^{-\ell} (1 - q^{\ell-j}) \right] \\ &= (-1)^{d-j} q^{-\binom{d+1}{2} - \binom{j}{2}} (q; q)_j (q; q)_{d-j}. \end{aligned}$$

Taking the reciprocal gives (77).

A.2 Why the inverse correction is beyond every Lambert order

From $h \sim z$ and endpoint flatness, $F'(h) = 2F(2h)$ is smaller than every power of h . The lower-Lambert relation $z = \lambda 2^{-\lambda}$ implies $z^a = \lambda^a e^{-aL\lambda}$, which is itself smaller than every power λ^{-N} . Therefore

$$F'(h) = \mathcal{O}(z^a) = o(\lambda^{-N})$$

for arbitrary $a, N > 0$, proving that the logarithmic correction in (27) cannot alter any coefficient in the algebraic $1/\lambda$ expansion.

A.3 Ordinary-point analyticity

If $p_m(x_0) \neq 0$ in (80), divide by p_m to obtain an analytic first-order system for the vector $(y, y', \dots, y^{(m-1)})$. Standard existence and uniqueness produce a convergent power series solution for any initial vector. Hence every C^m solution of the equation is analytic near x_0 . This is the only differential-equation fact needed in [theorems 9.2](#) and [9.4](#).

B Explicit Appell polynomials through degree eight

Put $Q = 4^r$. In addition to (69),

$$\begin{aligned}\mathcal{A}_7(x; Q) &= x^7 - \frac{7}{3}Qx^5 + \frac{217}{135}Q^2x^3 - \frac{2347}{8505}Q^3x, \\ \mathcal{A}_8(x; Q) &= x^8 - \frac{28}{9}Qx^6 + \frac{434}{135}Q^2x^4 - \frac{9388}{8505}Q^3x^2 + \frac{1904369}{32531625}Q^4.\end{aligned}$$

The unreduced exact forms emitted by SymPy are

$$\begin{aligned}\mathcal{A}_6(x; Q) &= \frac{-2347Q^3 + 41013Q^2x^2 - 99225Qx^4 + 59535x^6}{59535}, \\ \mathcal{A}_7(x; Q) &= \frac{x(-2347Q^3 + 13671Q^2x^2 - 19845Qx^4 + 8505x^6)}{8505}, \\ \mathcal{A}_8(x; Q) &= \frac{1904369Q^4 - 35909100Q^3x^2 + 104583150Q^2x^4}{32531625} \\ &\quad + \frac{-101209500Qx^6 + 32531625x^8}{32531625}.\end{aligned}$$

C Reproducibility files

The ZIP archive contains:

<code>fabius_frontier_report.tex</code>	Complete report source.
<code>fabius_frontier_report.pdf</code>	Compiled and visually inspected report.
<code>frontier_experiments.py</code>	Fully commented symbolic/numerical program.
<code>corpus_manifest.txt</code>	Recursive list of all fifty-seven audited TeX paths and the exclusion-search protocol.
<code>results/numerical_results.txt</code>	Human-readable ledger of parameters and outputs.
<code>results/inverse_defect.csv</code>	Machine-readable inverse-defect table.
<code>results/reproduction_errors.csv</code>	Machine-readable cardinal residual table.
<code>figures/*.png, figures/*.pdf</code>	Source figures in raster and vector form.
<code>README.txt</code>	Build and rerun instructions.

The program can be rerun with

```
python frontier_experiments.py
```

and the report can be rebuilt with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error fabius_frontier_report.tex
```

D Complete recursive TeX manifest

```
Recursive TeX corpus manifest
Repository: VladimirReshetnikov/ProveIt
Root: Analysis/FabiusFunction/docs
Snapshot audited: main branch, 2026-08-27

Method note
-----
Every .tex path exposed by the recursive documentation tree was inventoried.
The living canonical expositions, the consolidated frontier, the Thue--Morse
Atlas, the distinct Fourier-decay drafts/audits, and all paper transcriptions
were inspected directly. Historical variants were checked against the
archive ledger and searched for the candidate themes of this report. The
archive itself states that byte-identical current-head copies were pruned and
that many entries are frozen historical versions; consequently duplicates
were mapped to their canonical topic rather than counted as independent
mathematical evidence.

A. Living and current-tree documents (25)
-----
1. FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2. Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3. Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4. fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5. non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6. non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
7. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/
  Rvachev_Up_Fourier_Decay.tex
8. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/
  Fourier_decay_of_Rvachev_up.tex
9. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/
  Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-
  infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.
  tex
13. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/
  rvachev_up_fourier_decay.tex
14. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/
  rvachev_up_fourier_decay.tex
15. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/
  Rvachev_Up_Fourier_Decay_Final.tex
16. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/
  Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/
  rvachev_up_fourier_decay_final.tex
18. non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/
  rvachev_up_fourier_decay_final.tex
19. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20. papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
21. papers/arXiv-1502.06738v2/Lacunary_products.tex
22. papers/arXiv-1702.05442v1/09-Function.tex
23. papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24. papers/arXiv-2211.13570v1/document.tex
25. papers/ubiq15/ubiq15-corrected.tex

B. Frozen archive documents (32)
-----
```


26. archive/early-notes/Fabius Asymptotic/Fabius Asymptotic.tex
27. archive/early-notes/K-fold summation over the signed Thue-Morse sequence/K-fold summation over the signed Thue-Morse sequence.tex
28. archive/standalone-studies/Non-Elementarity_of_the_Fabius_Function/Non-Elementarity_of_the_Fabius_Function.tex
29. archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30. archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
31. archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33. archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34. archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35. archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36. archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
37. archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38. archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
39. archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40. archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41. archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42. archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
43. archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44. archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
45. archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46. archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47. archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
48. archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50. archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
51. archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52. archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
53. archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
54. archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex
55. archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56. archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57. archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

Candidate-topic exclusion searches

The living consolidated frontier, Atlas, all twelve Fourier-decay documents, and all seven paper transcriptions were searched for the following phrases and close variants: Strang--Fix, polynomial reproduction, cardinal reproduction, inverse-moment Appell, D-finite, holonomic, P-recursive, and the continuous identity $F(1/2+h)=1/2+2h\cdot F(h)$. No matching treatment of the report's three principal branches was found. Existing q-Richardson formulas, zero-counting results, endpoint Lambert-W expansions, Bell-polynomial saddle algebra, exact dyadic arithmetic, and finite-approximant midpoint identities are treated as prior inputs and are not claimed as new here.

References

- [1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction*, GitHub repository, <https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction>, repository snapshot audited 27 August 2026.
- [2] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex, 2026.
- [3] V. Reshetnikov, *Non-formalized research frontiers for the Fabius function and Rvachev's up-function*, Analysis/FabiusFunction/docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex, 2026.

- [4] V. Reshetnikov, *Thue–Morse Formula Atlas*, Analysis/FabiusFunction/docs/non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex, 2026.
- [5] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, arXiv:1702.05442, 2017.
- [6] J. Arias de Reyna, *Arithmetic of the Fabius function*, arXiv:1702.06487, 2017.
- [7] N. Dyn and A. Ron, *Multiresolution analysis by infinitely differentiable compactly supported functions*, Applied and Computational Harmonic Analysis **2** (1995), 15–20.
- [8] G. Strang and G. Fix, *A Fourier analysis of the finite element variational method*, in *Constructive Aspects of Functional Analysis*, Edizioni Cremonese, Rome, 1973, 795–840.
- [9] S. Li and Y. Shen, *Pseudo box splines*, Applied and Computational Harmonic Analysis **26** (2009), 344–356, DOI: 10.1016/j.acha.2008.07.004.
- [10] R. P. Stanley, *Differentiably finite power series*, European Journal of Combinatorics **1** (1980), 175–188.
- [11] R. M. Corless, G. H. Gonnet, D. E. G. Hare, D. J. Jeffrey, and D. E. Knuth, *On the Lambert W function*, Advances in Computational Mathematics **5** (1996), 329–359.
- [12] S. Roman, *The Umbral Calculus*, Academic Press, 1984.
- [13] L. Comtet, *Advanced Combinatorics: The Art of Finite and Infinite Expansions*, Reidel, 1974.
- [14] J.-P. Allouche and J. Shallit, *Automatic Sequences: Theory, Applications, Generalizations*, Cambridge University Press, 2003.
- [15] E. Prouhet, *Memoire sur quelques relations entre les puissances des nombres*, C. R. Acad. Sci. Paris **33** (1851), 225.